

XR-500 Industrial Controller

Configuration & API Reference

Document ID: DOC-XR500-API-v1.4

Protocol Version: REST API v2 / Modbus-TCP v1.1

1. Communication Settings & Interface Protocols

1.1 Network Configuration

The XR-500 system provides dual Ethernet interfaces (ETH1, ETH2) supporting standard industrial communications including Modbus/TCP, EtherNet/IP, and a native RESTful JSON API.

- **Default Interface (ETH1):** Static IP 192.168.1.100
- **Default Port Settings:**
 - Web GUI / REST API: Port 80 / 443
 - Modbus/TCP: Port 502
- **Network Timeout Setting:**
 - Parameter NET_TIMEOUT_MS: Default 3000 ms (Allowed Range: 500 – 10000 ms)
 - Network Retry Attempts (NET_RETRY_CNT): Default 3 (Allowed Range: 1 – 5)

2. System Configuration Parameters

2.1 Configurable Register Map

The following table outlines key internal configuration parameters accessible via Modbus holding registers or REST API endpoints.

Register	Parameter Name	Data Type	Default Value	Allowed Range	Unit	Description
40001	SYS_MODE	UINT16	1	0 – 3	—	0=Standby, 1=Auto, 2=Manual, 3=Cal
40005	PRESS_MAX_LIMIT	FLOAT32	8.0	1.0 – 12.0	bar	High pressure trip threshold

40012	FLOW_TARGET	FLOAT32	80.0	0.0 – 180.0	L/min	Operational target flow rate
40020	TEMP_TRIP_SET	FLOAT32	60.0	20.0 – 75.0	°C	Upper temperature emergency shutdown threshold
40033	COM_TIMEOUT	UINT16	5000	1000 – 30000	ms	API communication keep-alive timeout

3. REST API Interface Examples

3.1 Get System Status

- **Request Endpoint:** GET /api/v2/system/status
- **Headers:** Authorization: Bearer <token>
- **Sample JSON Response:**

JSON

```

1. {
2.   "device_id": "XR500-88492",
3.   "firmware_version": "v2.4.1",
4.   "system_state": "RUNNING",
5.   "telemetry": {
6.     "operating_pressure_bar": 4.2,
7.     "flow_rate_lpm": 85.0,
8.     "internal_temp_c": 38.5,
9.     "supply_voltage_vdc": 24.1
10.  },
11.   "status_code": 200
12. }
```

3.2 Update Target Flow Rate

- **Request Endpoint:** POST /api/v2/config/parameters

- **Payload:**

JSON

```
13. {
14.   "parameter": "FLOW_TARGET",
15.   "value": 110.0
16. }
```

- **Response:** 200 OK on success, 400 Bad Request if value exceeds range.

4. Error Codes & Fault Diagnostics

Error Code	Hex Code	Description	Recommended Action
ERR_101	0x0065	Supply Voltage Low (< 18.0 VDC)	Check DC power supply output voltage.
ERR_102	0x0066	Over-Pressure Limit Triggered	Verify discharge line valves and reset register PRESS_MAX_LIMIT .
ERR_204	0x00CC	Comm Timeout Limit Exceeded	Inspect cabling; confirm client matches COM_TIMEOUT setting.

17.